WORKED SOLUTION

STEP III 1992, Question 15

Probability and Statistics • Geometric Probability

STEP III 1992, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q15

The key idea

Because the goat's position is uniform over the field, every probability is just an area divided by a^2. So the whole question comes down to finding the area of the part of the square within distance r of the corner O. Once r passes a, the quarter-circle spills out of the square, and the clean way to find the area left inside is to cut it into two right-angled triangles and a circular sector. The density then follows by differentiating, and the tethered goat is the same idea with a different region.

Solution

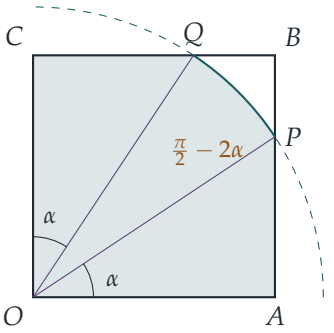
Put O at the origin with OA along the x-axis and OC along the y-axis, so that the field is $0 \le x \le a$, $0 \le y \le a$. Write $F(r) = \mathrm{P}(R < r)$, which is the area of the part of the field with $x^2 + y^2 < r^2$, divided by a^2.

The distribution of R

When $0 < r \le a$. The quarter-disc of radius r centred at O lies entirely inside the field, so

$$F(r) = \frac{\frac{1}{4}\pi r^2}{a^2} = \frac{\pi r^2}{4a^2}.$$

When $a \le r \le a\sqrt{2}$. Now the arc meets the side AB (the line $x = a$) at $P = (a, \sqrt{r^2 - a^2})$ and the side CB (the line $y = a$) at $Q = (\sqrt{r^2 - a^2}, a)$. Let $\alpha = \cos^{-1}(a/r)$, which is the angle AOP (since $\cos \angle AOP = a/r$). By symmetry, angle COQ is also α.



The shaded region (the part of the field within distance r of O) splits into three pieces:

- triangle OAP, a right angle at A, with legs a and $\sqrt{r^2 - a^2}$, so area $\frac{1}{2}a\sqrt{r^2 - a^2}$;
- triangle OCQ, congruent to it, with the same area;
- the sector OPQ of radius r and angle $\frac{\pi}{2} - 2\alpha$, with area $\frac{1}{2}r^2\left(\frac{\pi}{2} - 2\alpha\right) = \frac{\pi r^2}{4} - r^2\alpha$.

Adding these and dividing by a^2,

$$\begin{aligned} F(r) &= \frac{a\sqrt{r^2 - a^2}}{a^2} + \frac{\pi r^2}{4a^2} - \frac{r^2}{a^2}\,\alpha \\ &= \left(\frac{r^2}{a^2} - 1\right)^{\frac{1}{2}} + \frac{\pi r^2}{4a^2} - \frac{r^2}{a^2}\cos^{-1}\left(\frac{a}{r}\right), \end{aligned}$$

which is the required result. (For $r \geq a\sqrt{2}$ the whole field is within distance r of O, so $F(r) = 1$. The formula gives $1 + \frac{\pi}{2} - 2 \cdot \frac{\pi}{4} = 1$ at $r = a\sqrt{2}$, as it should.)

The median

At $r = a$ we have $F(a) = \pi/4 \approx 0.785 > \frac{1}{2}$, so the median m lies in the range $0 < m < a$, where the simple formula holds:

$$\frac{\pi m^2}{4a^2} = \frac{1}{2} \quad \Longrightarrow \quad m = a\sqrt{\frac{2}{\pi}} \approx 0.798a.$$

The probability density function

The density is $f(r) = F'(r)$. For $0 < r < a$ this is $f(r) = \dfrac{\pi r}{2a^2}$.

For $a < r < a\sqrt{2}$ we need the derivative of $\cos^{-1}(a/r)$:

$$\frac{\mathrm{d}}{\mathrm{d}r}\cos^{-1}\left(\frac{a}{r}\right) = -\frac{1}{\sqrt{1 - a^2/r^2}} \cdot \left(-\frac{a}{r^2}\right) = \frac{a}{r\sqrt{r^2 - a^2}}.$$

Then, differentiating each term of F (using the product rule on the last),

$$\begin{aligned} f(r) &= \frac{r}{a\sqrt{r^2-a^2}} + \frac{\pi r}{2a^2} - \frac{2r}{a^2}\cos^{-1}\left(\frac{a}{r}\right) - \frac{r^2}{a^2}\cdot\frac{a}{r\sqrt{r^2-a^2}} \\ &= \frac{r}{a^2}\left(\frac{\pi}{2} - 2\cos^{-1}\left(\frac{a}{r}\right)\right), \end{aligned}$$

since the first and last terms cancel. So

$$f(r) = \begin{cases} \dfrac{\pi r}{2a^2}, & 0 < r \le a, \\ \dfrac{r}{a^2}\left(\dfrac{\pi}{2} - 2\cos^{-1}\left(\dfrac{a}{r}\right)\right), & a < r \le a\sqrt{2}, \\ 0, & \text{otherwise.} \end{cases}$$

This makes geometric sense: $f(r)\,\mathrm{d}r$ is (arc length of the circle of radius r inside the field) $\times\, \mathrm{d}r/a^2$, and that arc subtends an angle $\frac{\pi}{2}$ when $r \le a$ and $\frac{\pi}{2} - 2\alpha$ when $r > a$. The two pieces agree at $r = a$ (both give $\pi/(2a)$), and f falls to 0 at $r = a\sqrt{2}$, where $\alpha = \pi/4$.

The tethered goat

Once tethered, the goat can reach only the points of the field with $R \le a$, the quarter-disc of radius a centred at O, which has area $\frac{1}{4}\pi a^2$. Its position is still uniform, now over this quarter-disc. That is exactly the original model conditioned on the event $R \le a$.

? **Ask yourself** Which coordinate measures the distance from the fence OC?

With OC along the y-axis, the distance from OC is x. So we need the area of the part of the quarter-disc with $x > a/2$:

$$\begin{aligned} \int_{a/2}^{a} \sqrt{a^2-x^2}\,\mathrm{d}x &= \left[\frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_{a/2}^{a} \\ &= \frac{\pi a^2}{4} - \left(\frac{\sqrt{3}\,a^2}{8} + \frac{\pi a^2}{12}\right) = \frac{\pi a^2}{6} - \frac{\sqrt{3}\,a^2}{8}. \end{aligned}$$

(Geometrically, this is a sector of angle $\pi/3$ minus a triangle with sides $a/2$ and $\sqrt{3}a/2$, because the line $x = a/2$ meets the arc at angle $\pi/3$ from OA.) Dividing by the area $\frac{1}{4}\pi a^2$,

$$\mathrm{P}\big(\text{distance from } OC > \tfrac{a}{2} \,\big|\, R \le a\big) = \frac{2}{3} - \frac{\sqrt{3}}{2\pi} \approx 0.391.$$

Checks

- Continuity: both formulas for F give $\pi/4$ at $r = a$, and $F(a\sqrt{2}) = 1$.

- A Monte Carlo simulation with 4×10^5 uniform points in the unit square agreed with $F(r)$ to about three decimal places at $r = 0.3, 0.7, 1.1, 1.25, 1.4$. It gave a sample median of 0.7986 against $\sqrt{2/\pi} = 0.7979$, and a conditional proportion of 0.3914 against $\frac{2}{3} - \frac{\sqrt{3}}{2\pi} = 0.3910$.
- The density integrates numerically to 1 over $[0, a\sqrt{2}]$, and matches a numerical derivative of F at several points.
- The tethered answer is less than $\frac{1}{2}$, as it should be: the quarter-disc is thinner far from OC.

Common slips

- Forgetting that F needs a different formula for $r > a$ when finding the median. It's worth checking $F(a) > \frac{1}{2}$ first, which shows the median lies in the easy range.
- Using the sector angle $\frac{\pi}{2} - \alpha$ instead of $\frac{\pi}{2} - 2\alpha$. Two corner pieces are cut off, one at each side.
- Dividing by a^2 in the tethered part. The conditional probability divides by the area of the quarter-disc, $\frac{1}{4}\pi a^2$.

Beyond STEP

Here R is a function of a point chosen uniformly in a region, and its distribution came from measuring the region where $R < r$. That is the general method for distributions of functions of random variables, and the density formula $f(r) = (\text{arc length})/a^2$ is a first taste of the "coarea" idea of slicing a region into level sets. The same geometric reasoning is behind problems such as Buffon's needle and the distance between two random points in a square. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both develop this geometric approach to continuous random variables.

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